

NFC-Based Smart Size Scanner Business Cards: A Flash Sales Strategy for the Indian Market

Business Research Document

May 9, 2026

1 Executive Summary

This document presents a business case for NFC-enabled smart size scanner business cards sold via flash sales in India. The product is a physical card containing an NFC tag that stores the user's body measurements and size profile. When tapped against a smartphone, the tag automatically opens a web app or brand integration, allowing instant sharing of size data with e-commerce platforms and fashion brands.

The primary revenue streams include direct tag sales (priced at 99), brand partnership fees, and affiliate commissions per size-scan transaction. With India's fashion e-commerce market growing rapidly and return rates on clothing exceeding 20–30%, reducing sizing uncertainty represents a significant value proposition for both consumers and brands.

2 Problem Statement

The Indian online fashion industry faces three interconnected challenges:

2.1 High Return Rates

Online apparel returns in India range between 20% and 35% of orders, with sizing issues accounting for approximately 60% of returns. This creates substantial logistical costs for brands and environmental waste across the supply chain.

2.2 Impulse Purchasing Without Size Context

Flash sales—characterized by limited-time offers on platforms such as Amazon, Flipkart, and Myntra—drive impulse purchases. Consumers frequently buy without verifying whether the product will fit, leading to disproportionate return rates compared to planned purchases.

2.3 No Persistent Size Memory Across Platforms

Consumers must manually enter measurements or size preferences on every platform they use. There is no unified, portable size profile that works across multiple e-commerce websites and brands. This friction leads to frustration, inaccurate purchases, and low brand loyalty.

3 Proposed Solution

The NFC Size Scanner Card is a physical business-card-sized device containing an NFC chip (NTAG213 or equivalent) with 144 bytes of usable user memory.

3.1 Product Concept

- A credit-card-sized card with an embedded NFC tag
- Pre-programmed with a unique URL linking to the user's size profile
- Tap to open: no app download required (uses browser-based NFC or cloud redirect)
- Profile includes: chest, waist, hip, inseam, shoe size, preferred fit (slim/regular/loose), and brand-specific size mappings

3.2 User Flow

1. User purchases the NFC card during a flash sale (99)
2. User taps the card against any NFC-enabled Android or iPhone (iOS 13+)
3. A web page opens prompting basic measurements (or pre-loaded via QR code at purchase)
4. User's size profile is stored in the cloud and associated with the NFC tag ID
5. When the user shops on a partner brand site and taps the card again, size data is shared via API to pre-fill size recommendations

4 Business Model

4.1 Revenue Streams

1. **Tag Sales:** Sale of NFC cards at 99 per unit via flash sales on Amazon India and Flipkart. Margin of approximately 30–40 per unit at scale.
2. **Brand Partnership Fees:** Brands pay a one-time integration fee (2–5 lakhs) plus per-transaction fees to access the size-scanning API. Estimate: 10 brand partnerships in Year 1 at 2 lakh each = 20 lakh.
3. **Affiliate Sizing Commissions:** Earn 15–25 per successful size-scan resulting in a purchase on a partner e-commerce platform. Projected 50,000 scans/month by end of Year 1.

5 Go-to-Market Strategy

5.1 Phase 1: Flash Sales Launch (Months 1–6)

- Partner with Amazon India and Flipkart for flash sale inventory allocation

- Limited-time offers: “First 5,000 buyers get 99 card (MRP 199)”
- Digital marketing campaign focusing on return-rate reduction messaging
- Unboxing and demo videos by fashion influencers with 100K–500K followings

5.2 Phase 2: D2C Brand Partnerships (Months 6–12)

- Direct outreach to direct-to-consumer (D2C) fashion brands (e.g., Bewakoof, Zivame, Limeroad)
- Offer white-label API integration so brands can embed size-scan functionality on their own checkout pages
- Co-branded cards sold through brand channels as loyalty/reward items

5.3 Phase 3: Scale and Retention (Year 2)

- Expansion to offline retail partners (Shoppers Stop, Westside, Reliance Trends)
- Subscription model for premium features (multiple profiles per family, export to tailors)
- International expansion to Southeast Asian markets

6 Technical Requirements

6.1 NFC Chip Specifications

- **Chip:** NXP NTAG213 or ST25TV (ISO 14443 Type 2 compatible)
- **User Memory:** 144 bytes (sufficient for a compressed size profile or a URL pointer)
- **Read Range:** 1–4 cm (proximity tap)
- **Write Cycles:** Minimum 100,000 endurance
- **Form Factor:** Inlay embedded in PVC card (85.6 mm × 54 mm × 0.76 mm)

6.2 App Integration

- Web-based NFC reader support (Chrome on Android, Safari on iOS 13+ with NFC SDK)
- Cloud database (Firebase/AWS) for user size profiles indexed by NFC tag UUID
- REST API for brand integrations with OAuth2 authentication
- Data format: JSON with schema for size profile (compatible with industry standard sizing)

6.3 Brand API Connections

- API endpoint: `api.sizescan.in/v1/match`
- Input: `tag_id`, `brand_id`, `product_category`
- Output: recommended size string (e.g., “M”, “32”, “UK 10”)
- Rate limiting: 100 requests/minute per brand
- SLA: 99.5% uptime, <200ms response time

7 Financial Projections

The following table summarizes the projected revenue and costs for Year 1 and Year 2 of operations.

7.1 Revenue Breakdown

Revenue Stream	Year 1	Year 2
Tag Sales (units)	60,000	240,000
Tag Sales Revenue (@ 99/unit)	59.4 lakh	237.6 lakh
Brand Partnership Fees (10 in Y1, 25 in Y2)	20 lakh	50 lakh
Affiliate Sizing Commissions (@ 20/scan)	12 lakh	48 lakh
Total Revenue	91.4 lakh	335.6 lakh

Table 1: Year 1 and Year 2 Revenue Projections

7.2 Cost Structure

Cost Item	Year 1	Year 2
NFC Tag Manufacturing (per unit: 25)	15 lakh	60 lakh
Cloud Infrastructure & API Hosting	3 lakh	8 lakh
Marketing & Advertising	20 lakh	45 lakh
Brand Integration Tech Support	5 lakh	12 lakh
Team & Operations	10 lakh	22 lakh
Total Costs	53 lakh	147 lakh

Table 2: Year 1 and Year 2 Cost Projections

7.3 Profitability Summary

8 Risk Analysis

8.1 Low Consumer Adoption of NFC-Based Size Scanning

Risk: Indian consumers may be unfamiliar with NFC interactions or skeptical of sharing body measurements. **Mitigation:** Heavy emphasis on ease-of-use (tap, no app download)

Metric	Year 1	Year 2
Total Revenue	91.4 lakh	335.6 lakh
Total Costs	53 lakh	147 lakh
Net Profit	38.4 lakh	188.6 lakh
Profit Margin	42%	56%

Table 3: Year 1 and Year 2 Profitability

in all marketing. First-time user education via video tutorials. Gamification of profile completion with discount rewards.

8.2 Technology Replacement Risk

Risk: NFC may be superseded by other technologies (e.g., Bluetooth LE, UWB) or smartphone manufacturers may restrict NFC functionality. **Mitigation:** Design the system to be technology-agnostic—the size profile is cloud-stored and accessible via QR code, URL, and future integration paths (Bluetooth beacon, voice assistant). NFC serves as an entry shortcut, not the sole access method.

8.3 Brand Resistance to Data Sharing

Risk: Fashion brands may be reluctant to integrate third-party size APIs due to competitive concerns or technical complexity. **Mitigation:** Position the API as a value-add that reduces returns and increases conversion rates—key metrics brands actively track. Offer white-label branding within the API response so the brand, not a third party, appears to the consumer.

9 Competitive Landscape

9.1 Direct Competitors

- **True Fit:** US-based B2B platform that uses algorithms to match consumers to sizes. Requires integration at the platform level, not a physical consumer product. No Indian market presence.
- **Myntra SizeID:** Myntra’s proprietary size recommendation tool integrated into its own app. Only works on Myntra and does not share data across brands. Not available as a standalone physical product.
- **ZealTime (India):** Emerging startup working on smart fitting solutions. Limited NFC integration and primarily B2B.

9.2 Differentiation Strategy

The NFC Size Scanner Card is the only product in the Indian market that combines:

1. A physical, portable NFC token that works across all brands and platforms
2. A flash sales distribution model that builds consumer awareness rapidly

3. A revenue model that combines hardware sales, brand partnerships, and affiliate commissions

10 Conclusion

The NFC-based smart size scanner represents a novel convergence of physical product marketing and cloud-based size data infrastructure. By targeting India's high return-rate fashion market with an affordable (99) NFC card distributed through flash sales, the product creates value for consumers (no more guessing sizes), brands (lower returns, higher conversion), and the platform itself (hardware sales + affiliate commissions + brand partnership fees).

With projected net profits of 38.4 lakh in Year 1 scaling to 188.6 lakh in Year 2, the business model demonstrates strong unit economics and multiple expansion levers.